

# Dishank Oza

+1 (669) 281-9480 | Santa Clara, CA | [dishankwork@gmail.com](mailto:dishankwork@gmail.com) | [linkedin.com/in/dishankkoza](https://linkedin.com/in/dishankkoza)

## SUMMARY

Software Engineer with an MS in Computer Science and 3+ years of full-stack development experience. Demonstrated expertise in building scalable distributed systems, event-driven AWS architectures, and AI-enabled cloud solutions driven by Spring Boot, React, and MCP tools.

## EDUCATION

Masters in Computer Science - Santa Clara University | 3.9/4.0

Sep 2024 - Jun 2026

Bachelors in Computer Engineering - University of Mumbai | 3.5/4.0

Aug 2017 - Jun 2021

Relevant Coursework: Distributed Computing, Machine Learning, Advance OS, Database Systems, Data Structures, Advanced Algorithms.

## PROFESSIONAL EXPERIENCE

### Teradata

Santa Clara, USA

#### Optimizer Engineering Intern II

Jun 2025 - Sept 2025

- Accelerated database compiler performance by migrating Teradata's **C++ compiler** from **LLVM 16 to 20** by removing typed pointers and deprecated APIs, and updating codegen/JIT interfaces to achieve a **20% reduction in instructions** and enhanced vectorization.
- Built a LLVM IR diffing and performance analysis tool for large-scale Teradata database system, debugging codegen across compiler versions, quantifying structural shifts, and generating visualizations, driving **~10% CPU-time reduction** on 1TB workloads.

### Freespace

Mumbai, India

#### Senior Software Engineer

May 2023 - Aug 2024

- Led a team of 4 engineers in agile sprints, delivering business-critical features on schedule; designed and implemented CI/CD pipelines with Jenkins, **reducing deployment cycles by 30%**. Recognized as **Freespace Captain for leadership** and delivery excellence.
- Eliminated 60%** of redundant computation costs by architecting a **Spring Boot**-based event-driven reservation microservice, using AWS DMS for CDC, Kinesis for **real-time streaming**, and Lambda with Step Functions to orchestrate just-in-time reservation execution.
- Redesigned concurrency model by replacing serialized database locks with **Redis distributed locks**, significantly improving throughput under load, reducing peak-time page load from **30s to 9s**, and resolving critical bottleneck transactions through SQL query profiling.
- Migrated synchronous booking system to an **event-driven architecture** using AWS SQS FIFO and Lambda, decoupling services for independent scaling and reducing peak-time server footprint from **12 to 8 instances (33% cost savings)**.

### Freespace

Mumbai, India

#### Software Engineer

Jun 2021 - May 2023

- Led the migration to a **React-based SPA** with React Router, lazy loading, route-level and custom Webpack code splitting, and reusable HOCs—reducing bundle size, doubling first-load performance, and improving repeat visit **speed by 100%**.
- Built a shared component library (React, TypeScript, Storybook) adopted across **3+ product teams**, establishing consistent UI patterns and accelerating feature delivery by **20% through reusable**, well-documented components
- Built a reusable serverless notification platform on AWS using **Lambda, SQS, SNS, and SES** with Serverless Framework for deployment, enabling scalable, fault-tolerant multi-channel communication (email, SMS, push notifications) across multiple products.
- Developed RESTful APIs using Spring Boot and GraphQL for an enterprise application serving **100,000+ users** across 30+ global clients, managing up to **5,000 concurrent** requests per minute.

## PROJECTS

### DevOps AI Agent (AWS Bedrock/Claude/GitHub/Slack) - [Link to Project](#)

- Built a multi-agent DevOps assistant on Amazon Bedrock using Bedrock Agents, Lambda MCP tools, GitHub, and Slack to automate incident triage, log analysis, RCA, and remediation planning with secure AuthN/AuthZ controls.

### AmberDB – Distributed SQL Database (Go/SQL) - [Link to Project](#)

- Designed and built a scalable, fault-tolerant SQL database with Raft-based replication, MVCC using Hybrid Logical Clocks, dynamic sharding, and 2PC for atomic cross-range transactions.

### SpaceLink (TypeScript/WebRTC/Next.js) - [Link to Project](#)

- Built a full-stack 2D virtual space platform where users customize interactive environments and engage in proximity-based video calls, using WebSockets with Protobuf for real-time avatar sync, WebRTC for media, and a Node.js/Next.js, PostgreSQL, and S3.

## TECHNICAL SKILLS

**Programming:** Java, Javascript, C++, Python, Go

**Backend/Frontend:** Spring Boot, Node.js, GraphQL, REST APIs, Microservices, React, Next.js, Webpack, Storybook, HTML/CSS

**AI/ML:** Ollama, MCP Tool Orchestration, Multi-Agent Workflows

**Databases:** PostgreSQL, Cassandra, MongoDB, Redis

**Cloud Technologies:** AWS(Lambda, SQS, SNS, SES, Kinesis, DMS, Step Functions, S3), Terraform, Docker, Kubernetes, Jenkins, Prometheus

**Certifications:** [AWS Certified Developer Associate](#), Scrum Alliance-Scrum Foundations

## ACHIEVEMENTS, AWARDS, AND LEADERSHIP

- Freespace Captain Award:** Recognized for engineering **leadership**, cross-team problem-solving, and on-time sprint delivery.
- SAE Aero Design 2020:** Core member of team ranked 9th globally and **2nd in Asia-Pacific among 150+ universities**.
- Smart India Hackathon Finalist:** Built an automated Raspberry Pi and OpenCV parking system; **placed 3rd in qualifiers**.
- Publication:** Authored the research paper "[Gesture-Based Interaction System](#)," published in Springer.